hex of the first 4 KB

C9200_9300_9400_95...

```
00000000  B1 0C 9C 0E C8 63 E8 81 2E C8 E9 84 55 55 C1 B3  ▒.£.└cΦü.└⌂äUU┴│
00000010  39 50 A1 43 39 B3 53 73 EA F3 02 7D 01 F4 D3 A2  9PíC9│Ss☺≤.}.⌠└ó
00000020  4F 28 05 F8 67 57 8E FA 60 58 64 74 5A 21 7E 5A  O(.°gWÄ·`XdtZ!~Z
00000030  B1 B2 72 57 00 00 00 00 DE AD BE EF DE AD BE EF  ▒▓rW....▌¡╜∩▌¡╜∩
00000040  00 00 04 0C 00 00 75 30 00 00 00 00 02 7D CE B8  ......u0.....}╬╕
00000050  31 1B 2E B8 00 00 00 00 00 00 00 00 00 00 00 00  1..╕............
00000060  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000070  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000080  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 03  ................
00000090  00 00 00 09 00 00 00 1D 4B 45 59 5F 54 4C 56 5F  ........KEY_TLV_
000000A0  50 41 43 4B 41 47 45 5F 43 4F 4D 50 41 54 49 42  PACKAGE_COMPATIB
000000B0  49 4C 49 54 59 00 00 00 00 00 00 09 00 00 00 0B  ILITY...........
000000C0  46 52 55 5F 52 50 5F 54 59 50 45 00 00 00 00 09  FRU_RP_TYPE.....
000000D0  00 00 00 18 4B 45 59 5F 54 4C 56 5F 50 41 43 4B  ....KEY_TLV_PACK
000000E0  41 47 45 5F 42 4F 4F 54 41 52 43 48 00 00 00 09  AGE_BOOTARCH....
000000F0  00 00 00 0E 41 52 43 48 5F 69 36 38 36 5F 54 59  ....ARCH_i686_TY
00000100  50 45 00 00 00 00 00 09 00 00 00 14 4B 45 59 5F  PE..........KEY_
00000110  54 4C 56 5F 42 4F 41 52 44 5F 43 4F 4D 50 41 54  TLV_BOARD_COMPAT
00000120  00 00 00 09 00 00 00 10 42 4F 41 52 44 5F 63 61  ........BOARD_ca
00000130  74 39 6B 5F 54 59 50 45 00 00 00 09 00 00 00 18  t9k_TYPE........
00000140  4B 45 59 5F 54 4C 56 5F 43 52 59 50 54 4F 5F 4B  KEY_TLV_CRYPTO_K
00000150  45 59 53 54 52 49 4E 47 00 00 00 09 00 00 00 0A  EYSTRING........
00000160  45 6E 43 72 59 70 54 69 4F 6E 00 00 00 00 00 09  EnCrYpTiOn......
00000170  00 00 00 0B 43 57 5F 42 45 47 49 4E 3D 24 24 00  ....CW_BEGIN=$$.
00000180  00 00 00 09 00 00 00 11 43 57 5F 46 41 4D 49 4C  ........CW_FAMIL
00000190  59 3D 24 63 61 74 39 6B 24 00 00 00 00 00 00 09  Y=$cat9k$.......
000001A0  00 00 00 27 43 57 5F 49 4D 41 47 45 3D 24 63 61  ...'CW_IMAGE=$ca
000001B0  74 39 6B 5F 69 6F 73 78 65 2E 31 36 2E 31 32 2E  t9k_iosxe.16.12.
000001C0  30 36 2E 53 50 41 2E 62 69 6E 24 00 00 00 00 09  06.SPA.bin$.....
000001D0  00 00 00 15 43 57 5F 56 45 52 53 49 4F 4E 3D 24  ....CW_VERSION=$
000001E0  31 36 2E 31 32 2E 30 36 24 00 00 00 00 00 00 09  16.12.06$.......
000001F0  00 00 00 37 43 57 5F 46 55 4C 4C 5F 56 45 52 53  ...7CW_FULL_VERS
00000200  49 4F 4E 3D 24 31 36 2E 31 32 2E 30 36 2E 30 2E  ION=$16.12.06.0.
00000210  36 33 35 33 2E 31 36 33 30 38 32 38 33 30 39 2E  6353.1630828309.
00000220  2E 47 69 62 72 61 6C 74 61 72 24 00 00 00 00 09  .Gibraltar$.....
00000230  00 00 00 34 43 57 5F 44 45 53 43 52 49 50 54 49  ...4CW_DESCRIPTI
```

```
00000240  4F 4E 3D 24 43 69 73 63 6F 20 49 4F 53 20 53 6F  ON=$Cisco IOS So
00000250  66 74 77 61 72 65 2C 20 49 4F 53 2D 58 45 20 53  ftware, IOS-XE S
00000260  6F 66 74 77 61 72 65 24 00 00 00 09 00 00 00 09  oftware$........
00000270  43 57 5F 45 4E 44 3D 24 24 00 00 00 00 00 00 0C  CW_END=$$.......
00000280  00 00 01 88 01 00 02 01 01 02 00 04 00 00 01 7C  ...ê...........│
00000290  03 00 61 04 00 28 43 4E 3D 43 69 73 63 6F 53 79  ..a..(CN=CiscoSy
000002A0  73 74 65 6D 73 3B 4F 55 3D 49 4F 53 2D 58 45 3B  stems;OU=IOS-XE;
000002B0  4F 3D 43 69 73 63 6F 53 79 73 74 65 6D 73 05 00  O=CiscoSystems..
000002C0  08 36 31 33 34 38 33 39 32 06 00 28 43 4E 3D 43  .61348392..(CN=C
000002D0  69 73 63 6F 53 79 73 74 65 6D 73 3B 4F 55 3D 49  iscoSystems;OU=I
000002E0  4F 53 2D 58 45 3B 4F 3D 43 69 73 63 6F 53 79 73  OS-XE;O=CiscoSys
000002F0  74 65 6D 73 07 00 01 00 08 00 01 01 09 00 01 00  tems............
00000300  0A 00 01 01 0B 01 00 53 70 B7 2B 34 8C C7 23 5B  .......Sp╗+4î╟#[
00000310  BD 54 EA BF F8 60 7C 68 C4 D2 A1 37 9F C3 E1 1D  ╝TΩ┐°`|h─╥í7ƒ├ß.
00000320  DD 58 C4 57 1C B2 02 29 88 4D 63 B5 84 BE D6 01  ▌X─W.▒.)êMc╡ä╛╓.
00000330  7A 13 86 95 2B 17 3C 3F 9D EB A1 A0 EA C3 AB 48  z.åò+.<?¥ðíáΩ├½H
00000340  34 9B E3 34 87 C6 DB 19 0A E3 18 6C 39 0B 39 8A  4¢π4ç╠╫..π.l9.9è
00000350  DF 88 B7 EA 82 D7 D8 B2 34 AC D3 CB 01 00 DB AF  ▀ê╗Ωé╫╪▒4¼╙╦..█»
00000360  3F 63 ED 6A 49 30 36 50 53 4C 2D ED C1 FF 1D 08  ?cφjI06PSL-φ┴ ..
00000370  4E B0 7B D0 A2 F6 C7 07 5A 2C 73 18 6E 0B 00 BC  N╲{╨¢ö÷╟.Z,s.n..╝
00000380  D5 25 03 1C 56 13 B0 AA 49 25 F4 6C 98 17 D9 97  ╒%..V.╲¬I%⌠lÿ.┘ù
00000390  82 EB E3 7F 58 7A AA D4 48 C5 E2 BB 7B E6 A0 FE  éδπ⌂Xz¬╘H╪╥╗{µá■
000003A0  19 97 70 68 AB 40 22 75 63 45 1D 71 FA 29 50 77  .ùph½@"ucE.q·)Pw
000003B0  F3 5F 85 F5 CB CF A1 A6 E4 0A 21 06 D9 9C C7 5A  ≤_à│╤╧íªΣ.!.┘£╟Z
000003C0  5C B6 0D D9 80 0C 6D A2 D5 7E 5B F6 C2 15 F7 43  \╢.┘Ç.mó╒~[÷┬.≈C
000003D0  95 06 3F DE 06 14 6B 83 46 C6 E7 1C 41 E1 20 16  ò.?▌..kâF╞τ.Aß .
000003E0  CA A8 2F 6F 04 73 3E 13 EF 14 78 F1 0B 9E 6B 4A  ╩¿/o.s>.∩.x±.₧kJ
000003F0  EA 65 05 85 89 DA 14 08 7E B4 3D BF 39 0B 9F 83  Ωe.àë┌..~┤=┐9.ƒâ
00000400  BC C1 E2 53 71 28 F8 0C 00 01 41 EB 27 C0 D3 B5  ╝┴┬Sq(°...Aδ'└╙╡
00000410  FB A2 0F DA 1F F8 AA 21 B9 43 BD A0 F2 D6 A3 99  √ó.┌.°¬!╣C╝á≥╓úÖ
00000420  B1 6E 6A 77 67 F2 50 4B C0 E1 E2 D7 90 34 3B 8D  ▒njwg≥PK└ßΓ╫É4;ì
00000430  C7 38 0E 1F D7 93 8E 02 66 AE 16 D5 23 47 82 DD  ╟8..╫ôÄ.f«.╒#Gé▌
00000440  00 00 00 00 DE AD BE EF DE AD BE EF 00 00 04 04  ....▌¡╜∩▌¡╜∩....
00000450  00 00 75 31 00 00 00 00 00 4E 97 74 02 7D CA AC  ..u1.....Nùt.}╩¼
00000460  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
```

```
00000470  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000480  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000490  00 00 00 00 00 00 00 00 00 00 00 03 00 00 00 09  ................
000004A0  00 00 00 1D 4B 45 59 5F 54 4C 56 5F 50 41 43 4B  ....KEY_TLV_PACK
000004B0  41 47 45 5F 43 4F 4D 50 41 54 49 42 49 4C 49 54  AGE_COMPATIBILIT
000004C0  59 00 00 00 00 00 00 00 09 00 00 00 0B 46 52 55 5F  Y...........FRU_
000004D0  52 50 5F 54 59 50 45 00 00 00 00 09 00 00 00 18  RP_TYPE.........
000004E0  4B 45 59 5F 54 4C 56 5F 50 41 43 4B 41 47 45 5F  KEY_TLV_PACKAGE_
000004F0  42 4F 4F 54 41 52 43 48 00 00 00 09 00 00 00 0E  BOOTARCH........
00000500  41 52 43 48 5F 69 36 38 36 5F 54 59 50 45 00 00  ARCH_i686_TYPE..
00000510  00 00 00 09 00 00 00 14 4B 45 59 5F 54 4C 56 5F  ........KEY_TLV_
00000520  42 4F 41 52 44 5F 43 4F 4D 50 41 54 00 00 00 09  BOARD_COMPAT....
00000530  00 00 00 10 42 4F 41 52 44 5F 63 61 74 39 6B 5F  ....BOARD_cat9k_
00000540  54 59 50 45 00 00 00 09 00 00 00 18 4B 45 59 5F  TYPE........KEY_
00000550  54 4C 56 5F 43 52 59 50 54 4F 5F 4B 45 59 53 54  TLV_CRYPTO_KEYST
00000560  52 49 4E 47 00 00 00 09 00 00 00 04 6E 6F 6E 65  RING........none
00000570  00 00 00 09 00 00 00 0B 43 57 5F 42 45 47 49 4E  ........CW_BEGIN
00000580  3D 24 24 00 00 00 00 09 00 00 00 11 43 57 5F 46  =$$.........CW_F
00000590  41 4D 49 4C 59 3D 24 63 61 74 39 6B 24 00 00 00  AMILY=$cat9k$...
000005A0  00 00 00 09 00 00 00 28 43 57 5F 49 4D 41 47 45  .......(CW_IMAGE
000005B0  3D 24 63 61 74 39 6B 2D 72 70 62 6F 6F 74 2E 31  =$cat9k-rpboot.1
000005C0  36 2E 31 32 2E 30 36 2E 53 50 41 2E 70 6B 67 24  6.12.06.SPA.pkg$
000005D0  00 00 00 09 00 00 00 15 43 57 5F 56 45 52 53 49  ........CW_VERSI
000005E0  4F 4E 3D 24 31 36 2E 31 32 2E 30 36 24 00 00 00  ON=$16.12.06$...
000005F0  00 00 00 09 00 00 00 37 43 57 5F 46 55 4C 4C 5F  .......7CW_FULL_
00000600  56 45 52 53 49 4F 4E 3D 24 31 36 2E 31 32 2E 30  VERSION=$16.12.0
00000610  36 2E 30 2E 36 33 35 33 2E 31 36 33 30 38 32 38  6.0.6353.1630828
00000620  33 30 39 2E 2E 47 69 62 72 61 6C 74 61 72 24 00  309..Gibraltar$.
00000630  00 00 00 09 00 00 00 34 43 57 5F 44 45 53 43 52  .......4CW_DESCR
00000640  49 50 54 49 4F 4E 3D 24 43 69 73 63 6F 20 49 4F  IPTION=$Cisco IO
00000650  53 20 53 6F 66 74 77 61 72 65 2C 20 49 4F 53 2D  S Software, IOS-
00000660  58 45 20 53 6F 66 74 77 61 72 65 24 00 00 00 09  XE Software$....
00000670  00 00 00 09 43 57 5F 45 4E 44 3D 24 24 00 00 00  ....CW_END=$$...
00000680  00 00 00 0C 00 00 01 88 01 00 02 01 01 02 00 04  .......ê........
00000690  00 00 01 7C 03 00 61 04 00 28 43 4E 3D 43 69 73  ...|..a..(CN=Cis
000006A0  63 6F 53 79 73 74 65 6D 73 3B 4F 55 3D 49 4F 53  coSystems;OU=IOS
```

```
000006B0  2D 58 45 3B 4F 3D 43 69 73 63 6F 53 79 73 74 65  -XE;O=CiscoSyste
000006C0  6D 73 05 00 08 36 31 33 34 38 32 37 33 06 00 28  ms...61348273..(
000006D0  43 4E 3D 43 69 73 63 6F 53 79 73 74 65 6D 73 3B  CN=CiscoSystems;
000006E0  4F 55 3D 49 4F 53 2D 58 45 3B 4F 3D 43 69 73 63  OU=IOS-XE;O=Cisc
000006F0  6F 53 79 73 74 65 6D 73 07 00 01 00 08 00 01 01  oSystems........
00000700  09 00 01 00 0A 00 01 01 0B 01 00 6A D4 31 EA D1  ...........j╚1Ω╥
00000710  9B FB 79 A3 D5 AE CF A4 E0 57 DE 79 EA E0 98 8E  ¢√yú╒«╧ñαW▐yΩαÿÄ
00000720  30 70 89 D4 F7 A5 0F C4 00 11 31 47 79 4A A2 39  0pë╚≈Ñ.─..1GyJó9
00000730  62 76 C8 B0 9C 56 45 F6 9C 4F D3 6C 73 3D 97 16  bv╙╲£VE÷£O╙ls=ù.
00000740  55 08 CC 2C DB 2A C7 D2 50 63 E7 B2 73 C8 E2 AC  U.╠,╛*╟╥Pcτ▒s╚Γ¼
00000750  69 6A 39 FA B0 D3 F0 C7 31 93 EB AE F4 53 BC 71  ij9·╲╙≡╢1ôδ«⌠S╝q
00000760  41 A7 8D 8D 4D 63 A4 D4 0F 08 19 50 A4 2D 52 EE  Aº¡¡Mcñ╘...Pñ-Rε
00000770  5C E8 AC 7E 34 67 77 6D 44 8D 7F 94 E6 E2 4F 57  \Φ¼~4gwmDì⌂öµΓOW
00000780  43 B4 DD 6A FE DF 51 24 7B AF D8 D5 09 7B 4A 5E  C┤▌j▪■Q${»╪╒.{J^
00000790  3E CD 23 B1 F5 27 CA E6 1C 10 3B 54 6A CB 87 6D  >═#╲│'╩µ..;Tj╦çm
000007A0  13 4C F7 6B 1F EB 20 32 5D 22 30 76 B1 C1 CC 9B  .L≈k.δ 2]"0v╱└╠¢
000007B0  FC 71 9C AC 48 FB 76 DF CE CB E5 4A 06 49 CB F1  ⁿq£¼H√v▄╙╥σJ.I╦±
000007C0  B6 AB C8 42 5B A7 36 4A EA 9A F2 95 9C 3E 47 22  ╢½╚B[º6JΩÚ≥ò£>G"
000007D0  7C E8 63 EB D9 59 28 F5 19 46 67 17 FA ED E2 E0  |Φcδ┘Y(│.Fg.·φΓα
000007E0  03 7D 7D 99 DD FC 2D C7 E5 EF 48 72 36 E6 9E 87  .}}Ö▌ⁿ-╟σ∩Hr6µ₧ç
000007F0  CC 6A 16 47 72 87 93 A6 10 30 FF FD ED 12 E9 E9  ╠j.Grçôª.0 ²φ.ΘΘ
00000800  B0 89 56 4E A5 B9 7C DB A5 24 67 0C 00 01 41 EB  ╲ëVNÑ╣│▄Ñ$g...Aδ
00000810  EA 05 00 C0 07 8C C8 8E D8 8E C0 8E D0 31 E4 FB  Ω..└.î╚Ä╪Ä└Ä╨1Σ√
00000820  FC BE 2D 00 AC 20 C0 74 09 B4 0E BB 07 00 CD 10  ⁿ╛-.¬ └t.┤.╗..═.
00000830  EB F2 31 C0 CD 16 CD 19 EA F0 FF 00 F0 55 73 65  δ≥1└═.═.Ω≡ .≡Use
00000840  20 61 20 62 6F 6F 74 20 6C 6F 61 64 65 72 2E 0D   a boot loader..
00000850  0A 0A 52 65 6D 6F 76 65 20 64 69 73 6B 20 61 6E  ..Remove disk an
00000860  64 20 70 72 65 73 73 20 61 6E 79 20 6B 65 79 20  d press any key 
00000870  74 6F 20 72 65 62 6F 6F 74 2E 2E 2E 0D 0A 00 00  to reboot.......
00000880  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000890  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000008A0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000008B0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000008C0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000008D0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000008E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
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000008F0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000900  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000910  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000920  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000930  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000940  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000950  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000960  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000970  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000980  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
00000990  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000009A0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000009B0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000009C0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000009D0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000009E0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000009F0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 FF  ................
00000A00  FF 1E 01 00 57 E5 04 00 00 00 FF FF 00 00 55 AA   ...Wσ....  ..U¬
00000A10  EB 66 48 64 72 53 0D 02 00 00 00 00 00 10 00 30  δfHdrS.....0
00000A20  00 01 00 80 00 00 10 00 00 00 00 00 00 00 00 00  ...Ç............
00000A30  00 00 00 00 80 61 00 00 00 00 00 00 FF FF FF 7F  ....Ça.....  ⌂
00000A40  00 00 20 00 01 15 13 00 FF 07 00 00 00 00 00 00  .. ..... .......
00000A50  00 00 00 00 00 00 00 00 6B 02 00 00 AB E9 4D 00  ........k...½ΘM.
00000A60  00 00 00 00 00 00 00 00 00 00 00 01 00 00 00 00  ................
00000A70  00 C0 2D 01 00 00 00 00 8C D8 8E C0 FC 8C D2 39  .└-.....î╪Ä└ⁿî╥9
00000A80  C2 89 E2 74 16 BA 80 5F F6 06 11 02 80 74 04 8B  ┬ë╥t.│Ç_÷...Çt.ï
00000A90  16 24 02 81 C2 00 04 73 02 31 D2 83 E2 FC 75 03  .$.ü┬..s.1╥â╥ⁿu.
00000AA0  BA FC FF 8E D0 66 0F B7 E2 FB 1E 68 9F 02 CB 66  │ⁿ Ä╨f.╖╥√.hƒ.╦f
00000AB0  81 3E 58 3C 55 AA 5A 5A 75 17 BF 60 3C B9 83 5F  ü>X<U¬ZZu.┐`<╣â_
00000AC0  66 31 C0 29 F9 C1 E9 02 F3 66 AB 66 E8 32 29 00  f1└)·┴⊖.≤f½fΦ2).
00000AD0  00 66 B8 D2 03 00 00 66 E8 E9 00 00 00 F4 EB FD  .f╕╥...fΦΘ...⌠δ²
00000AE0  38 06 FB 02 74 05 A2 FB 02 EB 00 66 9C 0F A0 0F  8.√.t.ó√.δ.f£.á.
00000AF0  A8 66 60 83 EC 2C 89 D6 89 E7 B9 0B 00 F3 66 A5  ¿f`â∞,ë╓τ╣..≤fÑ
00000B00  66 61 0F A9 0F A1 07 1F 66 9D CD 00 66 9C 1E 06  fa.⌐.í..f¥═.f£..
00000B10  0F A0 0F A8 66 60 FC 66 0F B7 E4 8C C8 8E D8 8E  .á.¿f`ⁿf.╖Σî╚Ä╪Ä
00000B20  C0 67 8B 7C 24 44 21 FF 74 08 89 E6 B9 0B 00 F3  └gï|$D! t.ëµ╣..≤
```

```
00000B30  66 A5 83 C4 2C 66 61 0F A9 0F A1 66 9D 66 C3 66  fÑâ─,fa.⌐.íf¥f╞f
00000B40  56 66 53 66 83 EC 2C 66 89 C3 66 83 F8 0A 75 0C  VfSfâ∞,fë├fâ°.u.
00000B50  66 B8 0D 00 00 00 66 E8 E3 FF FF FF 66 89 E0 66  f╕....fΦπ   fëαf
00000B60  E8 11 15 00 00 67 C7 44 24 10 07 00 67 C7 44 24  Φ....g╢D$...g╢D$
00000B70  18 01 00 67 C6 44 24 1D 0E 67 88 5C 24 1C 66 31  ...g╞D$..gê\$.f1
00000B80  C9 66 89 E2 66 B8 10 00 00 00 66 E8 50 FF FF FF  ╔fë┬f╕....fΦP
00000B90  66 8B 16 60 5F 66 85 D2 74 22 66 B9 FF FF 00 00  fï.`_fà╥t"f╣  ..
00000BA0  66 89 D6 66 83 C2 05 EC A8 20 74 08 88 D8 66 89  fë╓fâ┬.∞¿ t.ê╪fë
00000BB0  F2 EE EB 08 66 49 74 F4 F3 90 EB EB 66 83 C4 2C  ≥εδ.fIt⌠≤Éδδfâ─,
00000BC0  66 5B 66 5E 66 C3 66 53 66 89 C3 67 66 0F BE 03  f[f^f├fSfë├gf.╝.
00000BD0  84 C0 74 0A 66 43 66 E8 63 FF FF FF EB ED 66 5B  ä└t.fCfΦc   δφf[
00000BE0  66 C3 07 4E 6F 20 73 65 74 75 70 20 73 69 67 6E  f├.No setup sign
00000BF0  61 74 75 72 65 20 66 6F 75 6E 64 2E 2E 2E 0A 00  ature found.....
00000C00  66 BA A1 86 01 00 66 B9 20 00 00 00 66 4A 74 1F  f│íå..f╣  ...fJt.
00000C10  E6 80 E4 64 3C FF 75 04 66 49 74 13 A8 01 74 06  µÇΣd< u.fIt.¿.t.
00000C20  E6 80 E4 60 EB E6 A8 02 75 E2 66 31 C0 66 C3 66  µÇΣ`δµ¿.u┌f1└f├f
00000C30  83 C8 FF 66 C3 66 53 66 31 D2 8E E2 66 83 CA FF  â╙ f├fSf1╥Ä┌fâ╩
00000C40  8E EA 64 66 8B 0E 00 02 66 89 CA 67 66 8D 1C 08  ÄΩdfï...fë╩gfì..
00000C50  66 39 DA 74 16 66 42 64 66 89 16 00 02 E6 80 65  f9┌t.fBdfë...µÇe
00000C60  66 A1 10 02 66 31 D0 74 E7 EB 03 66 31 C0 64 66  fí..f1╨tτδ.f1└df
00000C70  89 0E 00 02 66 5B 66 C3 66 56 66 53 66 83 EC 2C  ë...f[f├fVfSfâ∞,
00000C80  66 BB 00 01 00 00 66 4B 0F 84 C2 00 66 B8 20 00  f╗....fK.ä┬.f╕ .
00000C90  00 00 66 E8 9D FF FF FF 66 85 C0 74 06 66 31 C0  ..fΦ¥   fà└t.f1└
00000CA0  E9 AF 00 66 89 E0 66 E8 CA 13 00 00 67 C7 44 24  Θ».fëαfΦ╩...g╢D$
00000CB0  1C 01 24 66 31 C9 66 89 E2 66 B8 15 00 00 00 66  ..$f1╔fë┬f╕....f
00000CC0  E8 1B FE FF FF 66 B8 20 00 00 00 66 E8 64 FF FF  Φ.■  f╕  ...fΦd
00000CD0  FF 66 85 C0 75 C7 66 E8 24 FF FF FF 66 89 C6 66   fà└u╟fΦ$   fë╞f
00000CE0  B8 20 00 00 00 66 E8 4A FF FF FF 66 85 C0 75 AD  ╕ ...fΦJ   fà└u¡
00000CF0  66 85 F6 74 21 E4 92 66 83 E0 FE 66 83 C8 02 E6  fà÷t!ΣÆfâα▪fâ╙.µ
00000D00  92 66 B8 00 00 20 00 66 E8 28 FF FF FF 66 85 C0  Æf╕.. .fΦ(   fà└
00000D10  0F 84 72 FF EB 87 66 E8 E4 FE FF FF B0 D1 E6 64  .är δçfΦΣ▪  ╲╤µd
00000D20  66 E8 DA FE FF FF B0 DF E6 60 66 E8 D0 FE FF FF  fΦ┌▪  ╲▀µ`fΦ╨▪
00000D30  B0 FF E6 64 66 E8 C6 FE FF FF 66 B8 00 00 20 00  ╲ µdfΦ╞▪  f╕.. .
00000D40  66 E8 EF FE FF FF 66 85 C0 74 AA E9 4F FF 66 83  fΦ∩▪  fà└t¬ΘO fâ
00000D50  C8 FF 66 83 C4 2C 66 5B 66 5E 66 C3 66 55 66 57  ╙ fâ─,f[f^f├fUfW
00000D60  66 56 66 53 66 83 EC 0C 67 66 89 54 24 04 66 85  fVfSfâ∞.gfëT$.fà
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C9200_9300_9400_95...
00000D70  C0 0F 84 D6 00 66 89 C2 66 83 E2 0F 66 C1 E8 04  └.ä╓.fë┬fâ╒.f┴Φ.
00000D80  8E E0 67 66 89 14 24 66 89 CF 66 31 D2 66 31 ED  Äαgfë.$fë╧f1╥f1φ
00000D90  66 83 C8 FF 67 66 8B 74 24 20 67 66 8D 5E FF 67  fâ╚ gfït$ gfì^ g
00000DA0  66 89 5C 24 08 67 66 8B 1C 24 64 67 8A 1B 84 DB  fë\$.gfï.$dgè.ä╙
00000DB0  0F 84 88 00 66 83 FA 02 74 3F 66 83 FA 03 74 40  .äê.fâ·.t?fâ·.t@
00000DC0  66 4A 74 0B 66 31 D2 80 FB 20 76 5E E9 82 00 66  fJt.f1╥Ç√ v^Θé.f
00000DD0  89 EE 80 FB 3D 75 08 67 80 3E 00 74 39 EB 08 66  ëεÇ√=u.gÇ>.t9δ.f
00000DE0  31 D2 80 FB 20 76 43 67 66 8D 6E 01 66 31 D2 67  1╥Ç√ vCgfìn.f1╥g
00000DF0  3A 1E 0F 95 C2 66 42 EB 31 80 FB 20 77 2C EB 27  :..ò┬fBδ1Ç√ w,δ'
00000E00  80 FB 20 76 22 67 66 39 44 24 08 7E 05 67 88 1F  Ç√ v"gf9D$.~.gê.
00000E10  66 47 66 40 EB 14 66 89 CF 66 89 F5 66 BA 03 00  fGf@δ.fë╧fë⌡f╢..
00000E20  00 00 66 31 C0 EB 03 66 31 D2 67 66 FF 04 24 67  ..f1└δ.f1╥gf .$g
00000E30  66 81 3C 24 00 00 01 00 0F 85 69 FF 67 66 83 7C  fü<$.....àï gfâ|
00000E40  24 20 00 74 1A 67 C6 07 00 EB 14 66 83 C8 FF EB  $ .t.g╞..δ.fâ╚ δ
00000E50  0E 67 66 8B 74 24 04 80 FB 3D 75 8B E9 78 FF 66  .gfït$.Ç√=uïΘx f
00000E60  83 C4 0C 66 5B 66 5E 66 5F 66 5D 66 C3 66 55 66  â─.f[f^f_f]f├fUf
00000E70  57 66 56 66 53 66 51 67 66 89 14 24 66 85 C0 0F  WfVfSfQgfë.$fà└.
00000E80  84 86 00 66 89 C7 66 83 E7 0F 66 C1 E8 04 8E E0  ääâ.fë╟fâτ.f┴Φ.Äα
00000E90  66 89 F9 66 31 D2 66 31 DB 66 31 C0 67 66 8D 71  fë·f1╥f1╛f1└gfìq
00000EA0  01 64 67 8A 09 66 89 F5 66 29 FD 66 83 FA 01 74  .dgè.fë⌡f)²fâ·.t
00000EB0  17 66 83 FA 02 74 38 84 C9 74 54 80 F9 20 76 38  .fâ·.t8ä╔tTÇ· v8
00000EC0  67 66 8B 1C 24 66 89 E8 67 8A 13 84 D2 75 0D 80  gfï.$fëΦgè.ä╥u.Ç
00000ED0  F9 20 76 3E 66 BA 02 00 00 00 EB 1F 84 C9 74 2F  · v>f╢....δ.ä╔t/
00000EE0  66 43 38 D1 0F 95 C2 66 0F B6 D2 66 42 EB 0C 84  fC8╤.ò┬f.╢╥fBδ.ä
00000EF0  C9 74 1C 80 F9 20 77 03 66 31 D2 66 89 F1 66 81  ╔t.Ç· w.f1╥fë±fü
00000F00  FE 00 00 01 00 75 95 EB 06 66 83 C8 FF EB 03 66  ■....uòδ.fâ╚ δ.f
00000F10  31 C0 66 5A 66 5B 66 5E 66 5F 66 5D 66 C3 56 57  1└fZf[f^f_f]f├VW
00000F20  89 C7 89 D6 51 C1 E9 02 F3 66 A5 59 83 E1 03 F3  ë╟ë╓Q┴Θ.≤fÑYâß.≤
00000F30  A4 5F 5E 66 C3 57 89 C7 66 0F B6 C2 66 69 C0 01  ñ_^f├Wë╟f.╢┬fi└.
00000F40  01 01 01 51 C1 E9 02 F3 66 AB 59 83 E1 03 F3 AA  ...Q┴Θ.≤f½Yâß.≤¬
00000F50  5F 66 C3 1E 0F A0 1F 66 E8 C1 FF FF FF 1F 66 C3  _f├..á.fΦ┴   .f├
00000F60  06 0F A0 07 66 E8 B4 FF FF FF 07 66 C3 66 55 66  ..á.fΦ┤   .f├fUf
00000F70  57 66 56 66 53 66 83 EC 10 67 66 8D 4C 24 04 67  WfVfSfâ∞.gfìL$.g
00000F80  66 8D 54 24 0C 67 66 8D 44 24 08 66 E8 F8 02 00  fìT$.gfìD$.fΦ°..
00000F90  00 67 66 8B 44 24 0C 67 66 39 44 24 08 0F 8D 99  .gfïD$.gf9D$..ìÖ
00000FA0  00 66 BA E0 2D 00 00 66 83 E8 40 74 2A 66 83 E8  .f╢α-..fâ°@t*fâ°
```

```
00000FB0  0F 75 06 66 B8 06 00 00 00 66 50 66 68 E7 2D 00  .u.f╕....fPfhτ-.
00000FC0  00 66 68 60 3C 00 00 66 E8 51 10 00 00 66 83 C4  .fh`<..fΦQ...fâ─
00000FD0  0C 66 BA 60 3C 00 00 66 52 66 68 ED 2D 00 00 66  .f╢`<..fRfhφ-..f
00000FE0  E8 53 10 00 00 67 66 8B 44 24 10 66 5E 66 5F 66  ΦS...gfïD$.f^f_f
00000FF0  BA E0 2D 00 00 66 83 F8 40 74 2A 66 83 F8 0F 75  ╢α-..fâ°@t*fâ°.u
00001000  06 66 B8 06 00 00 00 66 50 66 68 E7 2D 00 00 66  .f╕....fPfhτ-..f
00001010  68 60 3C 00 00 66 E8 03 10 00 00 66 83 C4 0C 66  h`<..fΦ....fâ─.f
00001020  BA 60 3C 00 00 66 52 66 68 0F 2F 00 00 66 E8 05  ╢`<..fRfh./..fΦ.
```

To understand the structural layout of the firmware image, the first 4 KB of the file were inspected using a hex editor.

At offset 0x00000090 a series of readable metadata entries becomes visible. These entries follow a key-value style structure and contain identifiers such as:

KEY_TLV_PACKAGE_COMPATIBILITY
FRU_RP_TYPE
KEY_TLV_PACKAGE_BOOTARCH
ARCH_i686_TYPE
KEY_TLV_BOARD_COMPAT
KEY_TLV_CRYPTO_KEYSTRING
CW_BEGIN
CW_FAMILY
CW_IMAGE
CW_VERSION
CW_FULL_VERSION
CW_DESCRIPTION
CW_END

The presence of these fields indicates that the firmware begins with a structured metadata block that describes the firmware package. The metadata includes information about hardware compatibility, architecture type, firmware version, and descriptive information about the image.

The block appears to follow a TLV-like (Type-Length-Value) encoding scheme where each metadata field contains an identifier and a corresponding value.

The metadata section terminates with the marker:

CW_END=$$

which marks the end of the firmware metadata header.

Immediately after this marker, a binary structure consistent with an X.509 certificate becomes visible. The certificate contains issuer information including references to Cisco Systems and IOS-XE, indicating that the firmware image is cryptographically signed.

Following the certificate region, executable boot code becomes visible, containing typical bootloader diagnostic strings such as:

"Use a boot loader"
"Remove disk and press any key to reboot"
"No setup signature found"

This indicates that the early portion of the firmware image contains the following high-level components:

1. Firmware metadata header (TLV-style key-value entries)
2. Embedded signing certificate
3. Bootloader executable code

Preliminary byte map of the firmware header:

Offset Range	Description
0x00000000–0x00000090	Initial header / unknown fields
0x00000090–0x00000270	Firmware metadata (TLV-style entries)
0x00000270–0x00000400	Embedded X.509 certificate
0x00000400–0x00001000	Bootloader / executable boot code